



Healthcare Innovation Challenge: Restore & Rehabilitate 2026

Sprout

AI-Powered Rehabilitation & Sobriety Companion

Rebuilding function, independence, and life — one day at a time.

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React · Vite · Node.js · Express · PostgreSQL · OpenAI · LangGraph

The Problem

48M

Americans with
Substance Use Disorder

SAMHSA, 2023

90%

Receive
No Treatment

NIDA, 2023

40–60%

Relapse Within
First Year

McLellan et al., 2000

Recovery is lonely between sessions.

People need daily accountability, personalized guidance, and low-barrier support — not just a crisis hotline.

Our Solution

A Tamagotchi-style rehabilitation companion powered by AI — transforming sobriety into a daily care loop.

Living Companion

Your pet's mood mirrors recovery progress. Positive actions bring joy — no punishment on difficult days.

AI Micro-Tasks

Mood check-ins are converted into tiny, realistic actions scaled to how the user feels right now.

Relapse-Risk Dashboard

Real-time risk score from mood, sleep, cravings, missed tasks, and journal sentiment.

Emergency Support Mode

High-risk language triggers support contacts and crisis resources — companion, not a clinical replacement.

Tech Stack



Frontend

- React 18 + Vite
- Tailwind CSS
- React Router DOM
- Fetch API / REST



Backend

- Node.js + Express
- PostgreSQL (pg)
- JWT Authentication
- bcryptjs + CORS



AI Layer

- OpenAI API (JSON mode)
- LangGraph stateful agent
- pgvector semantic memory
- Mood → risk classification

AI Architecture

Stateful LangGraph Agent – not a chatbot, a personalized rehabilitation engine



```
AI Output → { "risk_level": "medium", "task": "Take a 5-min walk + text a friend",  
  "pet_mood_change": "+8" }
```

Impact & Scientific Basis



25%

Target relapse
reduction (90-day)



Free

Mobile-first, no
cost barrier to access



iOS+Web

Single codebase,
global deployment

Evidence Base

- Digital CBT interventions reduce substance use relapse by 15-25% [Carroll et al., JAMA Psychiatry, 2014]
- Habit-loop apps improve medication adherence by 34% [Cafazzo et al., J Med Internet Res, 2012]
- AI ecological momentary intervention significantly reduces craving episodes [Gustafson et al., JAMA Psychiatry, 2014]

Ethics & Feasibility

Ethical Safeguards

Not a Clinical Replacement

Always refers users to professionals & crisis lines

AI Bias Mitigation

No diagnostic claims; plain-language risk scores

Privacy & Consent

JWT auth, opt-in data use, HIPAA-aligned architecture

Shame-Free Design

No punishment for relapse; companion re-engages gently

Phased Rollout

1

Phase 1 (0-3 mo)

MVP: check-ins, AI tasks, companion — pilot with 50 users

2

Phase 2 (3-6 mo)

Refine AI on real data; add journal analysis & risk dashboard

3

Phase 3 (6-12 mo)

Partner with outpatient clinics; HIPAA review; app-store launch

4

Phase 4 (12+ mo)

Payer integration, outcome reporting, scale to 10k+ users

References

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Sprout

"We built a mobile rehabilitation companion that turns sobriety goals into a Tamagotchi-style care loop.

AI personalizes daily micro-tasks, detects high-risk moments, and rebuilds independence — one day at a time."

Evidence-based · Ethical by design · Globally scalable



Let's Build This